Symmetrical Tree

AlgoExpert · Medium · Binary Trees

The problem

- Is the tree a mirror image of itself around its root?

Intuition

- A tree is symmetric if the left subtree mirrors the right.

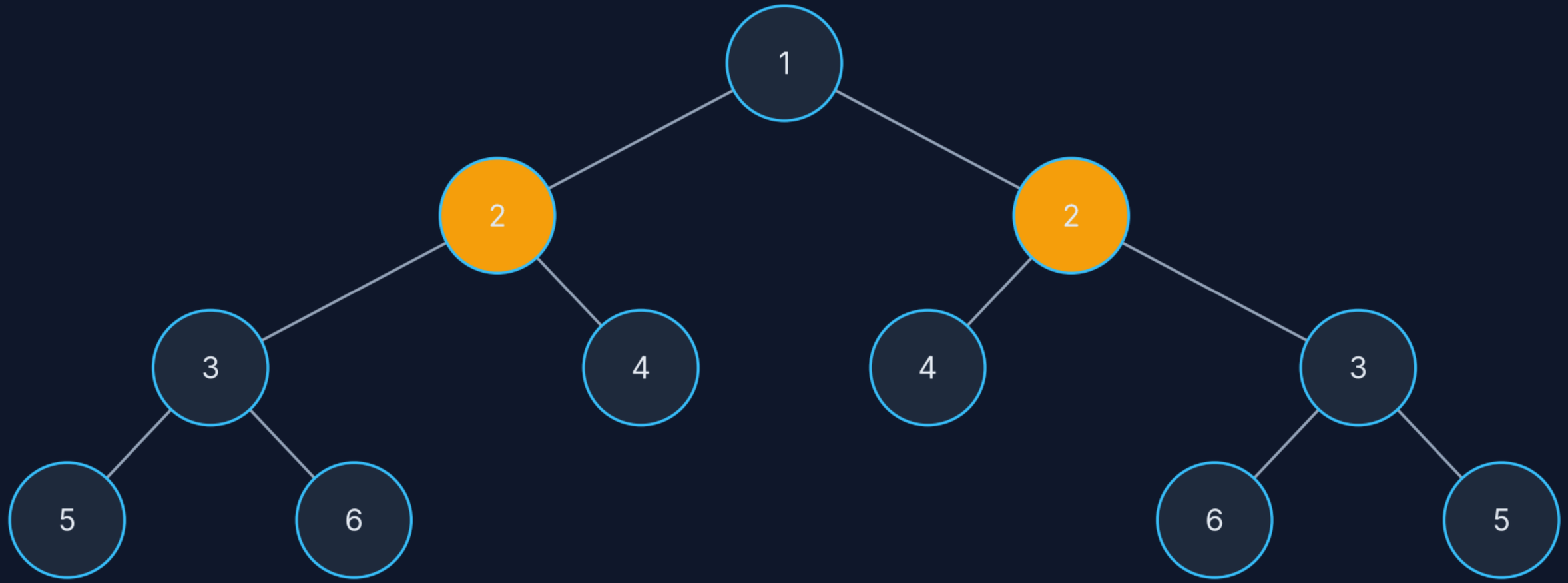
Brute force vs optimal

- Build the mirror tree and compare → O(n) time, O(n) space
- Recursive pairwise mirror check → O(n), O(h) space
- Or BFS checking each level is a palindrome

Approach

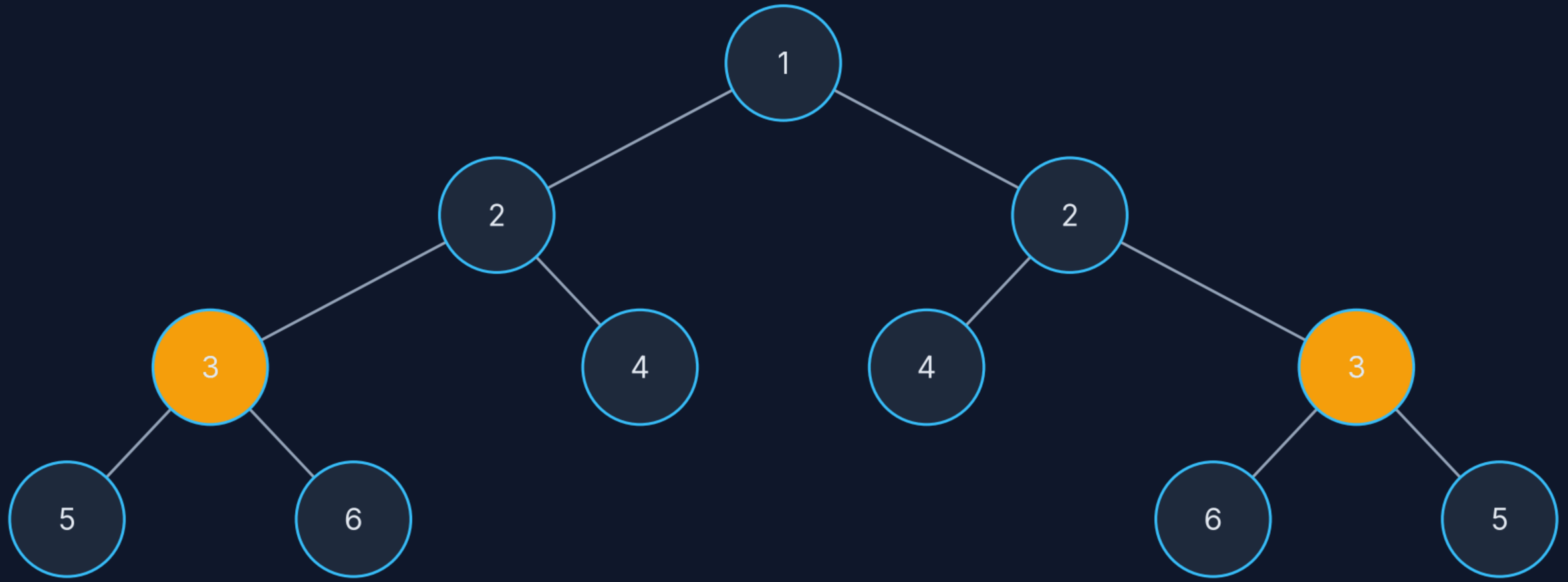
- 1. Compare two nodes at a time
- 2. Values must match
- 3. Outer children pair together and inner children pair together

Worked example



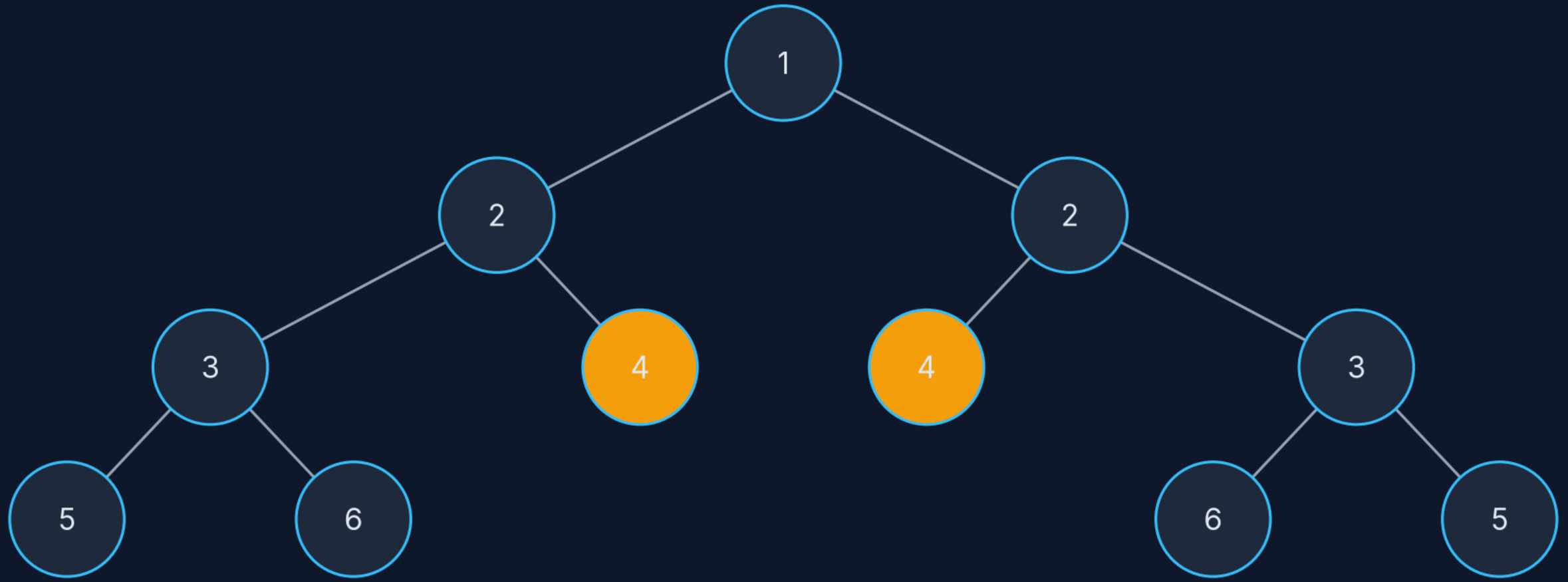
Compare 2 (left) with 2 (right) ✓

Worked example



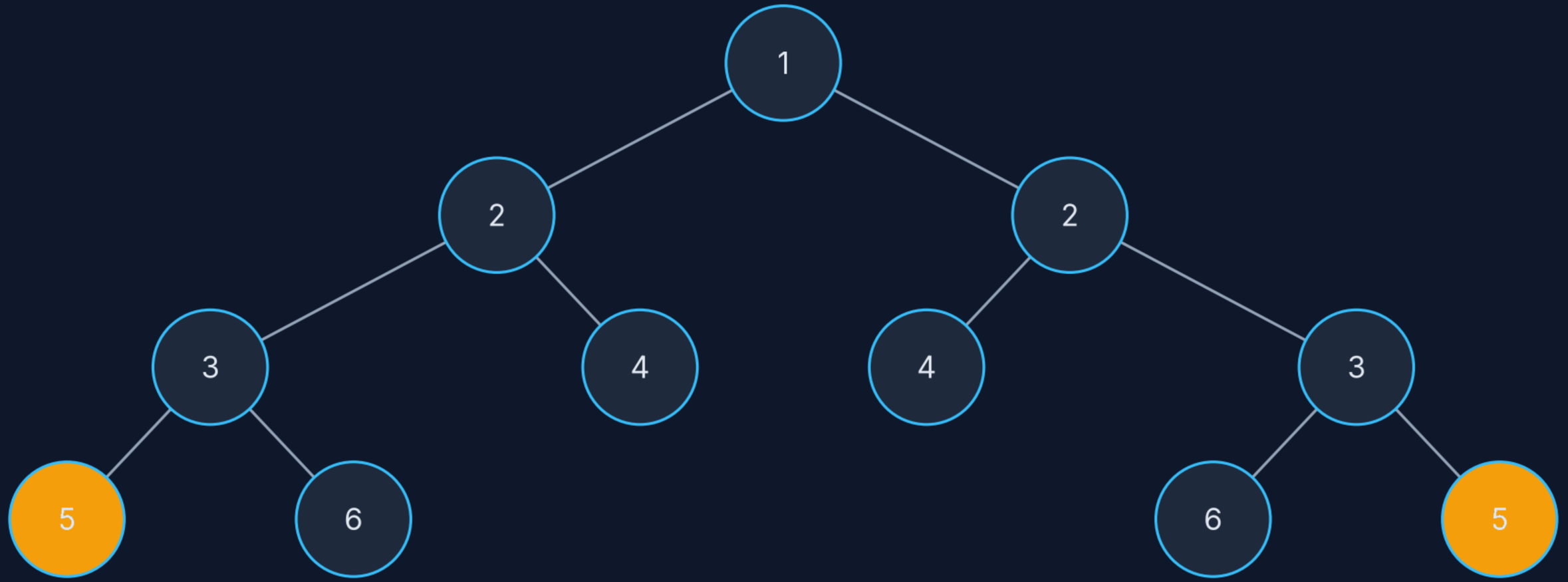
Compare 3 (left) with 3 (right) ✓

Worked example



Compare 4 (left) with 4 (right) ✓

Worked example



Compare 5 (left) with 5 (right) ✓

Java solution

```java
public class SymmetricalTree {
    public static boolean symmetricalTree(BinaryTree t) { return t == null || mirror(t.left, t.right); }

    private static boolean mirror(BinaryTree a, BinaryTree b) {
        if (a == null || b == null) return a == b;
        return a.value == b.value && mirror(a.left, b.right) && mirror(a.right, b.left);
    }
}
```

Complexity

- Time: O(n)
- Space: O(h)

Edge cases & pitfalls

- One side null, the other not → false
- Values must match, not just shapes
- Empty tree → symmetric

Interview tips & follow-ups

- Iterative with a queue of pairs
- Relation to Invert Binary Tree

Recap

- A tree is symmetric if the left subtree mirrors the right.
- Complexity: O(n)